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**GE Medical
Systems**

Technical Publication

**Direction 2253822 FMI
Revision 0**

GE Medical Systems FMI 25261 LightSpeed 1.1 Z-Axis Tracking Release

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Revision History

Rev	Date	Reason for Change
Rev. 0,	Date: 11-16-99	Initial Release

List of effective pages

Page(s)	Rev.
Title Page	0
Restricted/History/LOEP	0
Pages 1 to ??	0
Rear	-

LightSpeed 1.1 Z-Axis Tracking Release

EFFECTIVITY:

All LightSpeed QX/i System manufactured before 12-1-99 with a Gantry Model Number of 2180482. Some of the systems near this manufacturing cut in date may have the new revision of Software loaded. If the system has LightSpeed QX/i Applications software Version 6.x, the FMI is not required. Otherwise, the FMI is required.

PREREQUISITE:

All Open LightSpeed FMI's must be installed on the system prior to this FMI.

FMI 25256 – LightSpeed QX/i 1.0 Production Software – Effectivity - All LightSpeed Systems

FMI 25259 – LightSpeed QX/i DCB Upgrade – Effectivity – LightSpeed systems manufactured before August 1999.

Training is required before starting this FMI.

PURPOSE:

The FMI provides the following:

- Reduction of x-ray dose the patient is exposed to during an exam
- Enables the use of a Retrospective Cardiac Gating option package with the LightSpeed.
- Cine is also available for use on the LightSpeed.

RELATED FMI'S:

The following FMI's must be complete:

- 25259 - LightSpeed DAS Control Board Upgrade
- 25256 - LightSpeed 1.0 M4 Forward Production Release

MANPOWER REQUIREMENTS:

Labor Estimates: 6 hours

Travel Estimates: 1 hour

Training Estimates: 5 hours

(Not to be charged to the FMI)

SPECIAL TOOLS & TEST EQUIPMENT:

Standard Field Engineer Tool Kit.

Poly and QA Phantoms that are on each site

FURNISHED MATERIALS:

FMI kit 2253822 consists of the following material and documentation:

Part Number	Description	Qty
2253822FMI	FMI 25261 Document	1
2260334	LightSpeed QX/I System Software CDROMs	1
2211221-100	LightSpeed System Service Manual General	1
2211222-100	LightSpeed System Service Manual Advanced	1
2211217-100	System Installation Manual General	1
2211218-100	System Installation Manual Advanced	1
2211225-100	FRU Manual	1
2211227-100	LightSpeed Schematics and Diagrams Manual	1
2211228-100	Functional Interconnects Manual Advanced	1
2211214-100	Technical Reference Manual	1
2211220-100	Load From Cold Manual	1
2141834-100	CT PM Procedures Manual	1
2226999	TP Tutorial CDROM	1
2255226-100	LightSpeed 1.1 Service Introduction Planner	1
2211223-100	LightSpeed QX/I Tips and Work Arounds	1
2259500	LightSpeed QX/I Language CDROM	1
2211219-100	LightSpeed Pre-Installation Manual	1

DISPOSAL AND SCRAP:

Not Required. Any materials should always be disposed of per normal regional/local requirements.

PROCEDURE:

Introduction

This FMI includes the following activities:

- FMI Training (LightSpeed QX/I Software Rev. 1.1, Course ID CTQXISSW11)
- Load From Cold
- System Calibrations
- IQ Checks.

SECTION 1.0 General Precautions

Use the Technical Publications provided in the FMI kit to perform the following tasks. The previous release information is not accurate for this release.

SECTION 2.0 Determine if CDROM is Good

Note: There are some CDROM Drives that may have problems during a Software Load. The problem usually is not apparent until after the Applications Software has started to load and the system is trying to complete the load process. By performing a scsistat command in a shell, one can tell if the CDROM is in the suspected population. Run the command by doing the following:

1. Go to the Service Desktop
2. Select UNIX shell in the ToolChest menu.
3. Type: scsistat <ENTER>
4. In the menu text, you will see Device 1 6. If it is a Toshiba XM-6201 with rev 1037 firmware, then it is possible that it could experience problems during the software load. If this CDROM is on the equipment and you have some of the following symptoms, you may need to order a new CDROM drive. The new CDROM drive part number is 2253371. Other CDROM Drives or other firmware revisions have not experienced problems.

Symptoms of a bad CDROM Drive during a software Load:

CDROM drive loads the 3 IRIX OS CD's successfully but when attempting to load the Application SW CD, device timeouts on sc1d6l0 (CDROM on SCSIbus 1 at SCSI ID 6) occur. The applications software load appears to hang, timeout messages appear in the install shell window and/or device timeout messages appear in the 'console' shell window and the CDROM access LED is on solid.

SECTION 3.0 Perform Load From Cold

After evaluating the CDROM drive, perform the following instructions:

Using the Load From Cold document, 2211220-100, and the software included in the FMI kit, perform a software load. Follow the procedures to load the software and restore the system state from MOD. A Flash Download is required. Do not perform Section 2.17.2 System Performance Checks until after calibrations.

SECTION 4.0 Perform Z-Slope Calibration

1. Go to the Exam RX desktop and select Scanner Utilities.
2. Under Scanner Utilities, select Z-slope Calibration
3. Perform the Z-slope Calibrations. The scanner will add warm up scans if needed. These are air scans.

SECTION 5.0 Perform a DAS Gain Calibration

4. Go to the Exam RX desktop and select Scanner Utilities.
5. Under Scanner Utilities, select DAS Gain Calibration
6. A message will advise you that there should be nothing in the X-Ray beam path. Clear any objects from in the gantry.
7. Perform the DAS Gain Calibrations. These are air scans.

Note: After DAS Gain Cals, a pop up box will prompt you to do FastCal or Detailed Cal. Do not do these yet.

SECTION 6.0 Perform a Collimator Calibration

8. Go to Exam RX desktop, if not already present, in Scanner Utilities.
9. Under Scanner Utilities, select Collimator Calibration.
10. A message will advise you that there should be nothing in the X-Ray beam path. Clear objects in gantry.
11. Perform the Collimator Calibrations. These are air scans.

Note: Collimator Calibration checks the collimator aperture width during the calibration process. If collimator aperture measurements are incorrect, the calibration will fail with error messages indicating that the calibration process completed unsuccessfully with aperture failures. Refer to Appendix A in the rear of this document to help troubleshoot the failure.

Note: After DAS Gain Cals, a pop up box will prompt you to do FastCal or Detailed Cal. Continue per instructions below to do the calibrations.

SECTION 7.0 Perform a Full Detailed Calibration

12. Go to Exam RX desktop, if not already present, in Scanner Utilities.
13. Under Scanner Utilities, select Detailed Calibration.
14. The first set of scans are air scans. There should be nothing in the X-Ray beam path.
15. After the Air Scans are complete, the calibration process will request that a series of phantoms be placed on the phantom holder to be centered and scanned. Follow the on screen prompts to center and perform the scans. There are 3 sets of phantoms that will be required for scanning, the 48 cm poly phantom, the 35 cm poly, and the water phantom.
16. When centering the water phantom, make sure the phantom is aligned over the water section, not the section of the phantom with the high density pieces.

SECTION 8.0 Perform CT N# Adjustments

17. Go to Exam RX desktop (the left monitor), if not already present, and select Scanner Utilities.
18. Under Scanner Utilities, select Auto CT N# Adjustments.
19. The process will require that the water phantom be scanned. It should be still centered from the last phantom scans. Follow the on screen prompts to center the phantom.
20. You may see an error message stating the CT N#'s should be adjusted no more than 30 minutes after performing scans that would have the system warm. If it has been more than 30 minutes since scanning, perform a FastCal before continuing.
21. Once the phantom is centered, the software will prompt scanning and adjust the N #'s automatically. This will continue until all N#'s are adjusted. If the software is unable to adjust a N#, the software will stop. N#'s will have to adjusted manually. Refer to the System Installation Manual, 2211218-100, System Tests Chapter to adjust CT N#'s.

SECTION 9.0 Perform IQ Checks

Using the information provided in the System Installation Manual, 2211218-100, Chapter 10 System Tests, perform the IQ checks.

SECTION 10.0 Save System State

Refer to the Load From Cold Manual, 2211220-100, and perform a Save System State. This will save the new calibrations for the system.

SECTION 11.0 System Checkout

Refer to the Load From Cold Manual, 2211220-100, and perform the system performance checks as described in the manual. Ensure all networking and filming processes are still functional.

SECTION 12.0 FMI Completion

Fill out and return the FMI Completion Certification Report paperwork (or electronically close your FMI dispatch) in accordance with your local procedure

Appendix A - Procedure to Check Collimator Calibration

1. After Collimator Calibration, Check and verify testAperture values are within specification"
2. Open a Unix Shell. Refer to ToolChest on Service Desktop.
3. Re-size the Unix Shell window to full width of monitor to better view the file.
4. Remote Shell into the SBC and view the tracking history file. See example below:

```
{ctuser@baye} rsh sbc <CR>
IRIX Release 6.5 IP32 baye_sbc
Copyright 1987-1999 Silicon Graphics, Inc. All Rights Reserved.
Last login: Tue Sep 21 14:11:09 CDT 1999 by ctuser@baye_oc
```

Remote Shell into the SBC

```
{ctuser@baye_sbc} cd /usr/g/service/state <CR>
{ctuser@baye_sbc}
```

Change Directories

```
{ctuser@baye_sbc} cat ssw.tracking.hist <CR>
```

View tracking history file

```
Mon Sep 18 13:33:53 2000 Collimator Cal: refAperture: 4.027080 testAperture: 2.502403 doseFactor: 0.621394
Mon Sep 18 13:33:53 2000 Collimator Cal: FET setting: 1
Mon Sep 18 13:33:53 2000 Collimator Cal: Tracking Target A: -4.153496 B: 4.683610
Mon Sep 18 13:33:53 2000 Collimator Cal: Polynomial ranges, A: 0.092441 0.961463 b: 0.055978 0.962666
Mon Sep 18 13:33:53 2000 Collimator Cal: Poly coeffs for A: 1.014603 -16.434570 20.909405 -14.573469 1.478881
Mon Sep 18 13:33:53 2000 Collimator Cal: Poly coeffs for B: -0.655768 15.490855 -18.642046 11.007548 0.546813
Mon Sep 18 13:34:04 2000 Collimator Cal: refAperture: 5.516100 testAperture: 3.602403 doseFactor: 0.653071
Mon Sep 18 13:34:04 2000 Collimator Cal: FET setting: 6
Mon Sep 18 13:34:04 2000 Collimator Cal: Tracking Target A: -7.766982 B: 7.496323
Mon Sep 18 13:34:04 2000 Collimator Cal: Polynomial ranges, A: 0.203643 0.910281 b: 0.149313 0.918530
Mon Sep 18 13:34:04 2000 Collimator Cal: Poly coeffs for A: -4.397934 4.508885 -25.636704 20.347289 -4.651806
Mon Sep 18 13:34:04 2000 Collimator Cal: Poly coeffs for B: 4.233526 1.697803 13.323312 -14.855839 5.632672
Mon Sep 18 13:34:16 2000 Collimator Cal: refAperture: 7.090450 testAperture: 4.972403 doseFactor: 0.701282
Mon Sep 18 13:34:16 2000 Collimator Cal: FET setting: 8
Mon Sep 18 13:34:16 2000 Collimator Cal: Tracking Target A: -11.651983 B: 11.645393
Mon Sep 18 13:34:16 2000 Collimator Cal: Polynomial ranges, A: 0.214761 0.921753 b: 0.147874 0.903242
Mon Sep 18 13:34:16 2000 Collimator Cal: Poly coeffs for A: -8.762280 5.978048 -28.342112 21.130280 -4.279839
Mon Sep 18 13:34:16 2000 Collimator Cal: Poly coeffs for B: 8.406672 4.520022 5.393836 -5.967171 1.866747
Mon Sep 18 13:34:27 2000 Collimator Cal: refAperture: 8.494130 testAperture: 6.362403 doseFactor: 0.749035
Mon Sep 18 13:34:27 2000 Collimator Cal: FET setting: 9
Mon Sep 18 13:34:27 2000 Collimator Cal: Tracking Target A: -15.521935 B: 15.931264
Mon Sep 18 13:34:27 2000 Collimator Cal: Polynomial ranges, A: 0.251317 0.900883 b: 0.172763 0.915170
Mon Sep 18 13:34:27 2000 Collimator Cal: Poly coeffs for A: -8.626588 -13.612431 35.238058 -62.212883 33.850375
Mon Sep 18 13:34:27 2000 Collimator Cal: Poly coeffs for B: 10.989647 2.171578 11.459196 -13.808160 5.622990
Mon Sep 18 13:34:40 2000 Collimator Cal: refAperture: 3.783280 testAperture: 2.302411 doseFactor: 0.608575
Mon Sep 18 13:34:40 2000 Collimator Cal: FET setting: 1
Mon Sep 18 13:34:40 2000 Collimator Cal: Tracking Target A: -4.494549 B: 5.607463
Mon Sep 18 13:34:40 2000 Collimator Cal: Polynomial ranges, A: 0.102290 0.949739 b: 0.070738 0.944092
Mon Sep 18 13:34:40 2000 Collimator Cal: Poly coeffs for A: -1.740428 -9.416504 9.778851 -6.461875 0.210345
Mon Sep 18 13:34:40 2000 Collimator Cal: Poly coeffs for B: 1.894391 9.521741 -11.524514 10.200300 -2.456051
Mon Sep 18 13:34:53 2000 Collimator Cal: refAperture: 5.101730 testAperture: 3.452411 doseFactor: 0.676714
Mon Sep 18 13:34:53 2000 Collimator Cal: FET setting: 6
Mon Sep 18 13:34:53 2000 Collimator Cal: Tracking Target A: -8.213212 B: 8.616191
Mon Sep 18 13:34:53 2000 Collimator Cal: Polynomial ranges, A: 0.140374 0.939111 b: 0.090676 0.918245
Mon Sep 18 13:34:53 2000 Collimator Cal: Poly coeffs for A: -5.552114 -7.173822 5.040774 0.865298 -3.461022
Mon Sep 18 13:34:53 2000 Collimator Cal: Poly coeffs for B: 5.812450 8.848028 -13.590513 12.291607 -3.253606
Mon Sep 18 13:35:03 2000 Collimator Cal: refAperture: 6.676240 testAperture: 5.072412 doseFactor: 0.759771
Mon Sep 18 13:35:03 2000 Collimator Cal: FET setting: 8
Mon Sep 18 13:35:03 2000 Collimator Cal: Tracking Target A: -13.011189 B: 13.307949
Mon Sep 18 13:35:03 2000 Collimator Cal: Polynomial ranges, A: 0.148425 0.951517 b: 0.114390 0.950093
Mon Sep 18 13:35:03 2000 Collimator Cal: Poly coeffs for A: -9.841308 -6.260733 1.125299 6.155735 -5.873017
Mon Sep 18 13:35:03 2000 Collimator Cal: Poly coeffs for B: 10.429165 6.902096 -6.496304 1.883261 1.945876
Mon Sep 18 13:35:13 2000 Collimator Cal: refAperture: 8.080040 testAperture: 6.012411 doseFactor: 0.744107
Mon Sep 18 13:35:13 2000 Collimator Cal: FET setting: 9
..... (truncated for clarity)
```

Scan 1 Lrg Spot / 4 x 1.25

Scan 2 Large Spot / 4 x 2.50

Scan 3 Large Spot / 4 x 3.75

Scan 4 Large Spot / 4 x 5.00

Scan 5 Small Spot / 4 x 1.25

Scan 6 Small Spot / 4 x 2.50

Scan 7 Small Spot / 4 x 3.75

Scan 8 Small Spot / 4 x 5.00

5. There are 8 lines which contain the fields **refAperture**, and **testAperture**
6. Verify the **testAperture** values against the following spec with respect to the value of the **refAperture** on the same line. There are many values listed! Carefully confirm that the correct values are matched to the specification.

RefAperture*	TestAperture** Specification	Scan	Focal Spot Size	Aperture
4.027080	2.141947 - 2.641947	1	Large	4 x 1.25
5.516100	3.333182 - 3.833182	2	Large	4 x 2.50
7.090450	4.705002 - 5.205002	3	Large	4 x 3.75
8.494130	6.064653 - 6.564653	4	Large	4 x 5.00
3.783280	1.894207 - 2.394207	5	Small	4 x 1.25
5.101730	3.184344 - 3.684344	6	Small	4 x 2.50
6.676240	4.812432 - 5.312432	7	Small	4 x 3.75
8.080040	5.80195 - 6.30195	8	Small	4 x 5.00

*refAperture value is the collimator aperture width in mm when the CAMs are set up in the non-tracking mode.

**testAperture value is the final aperture width as a result of Tracking Mode algorithm.

If testAperture is not within Spec:

1) Stop! Do not continue with Detail Cal.

In only one incident, a LightSpeed detector was replaced due to this type of error, but DAS noise, particularly on the Z-Channels, may cause this error.

2) Save Scan Data (8 scanfiles from Collimator Cal) on MOD

3) Follow Tracking troubleshooting flow diagram starting at Flow 'D' and record test results. Details on tracking theory and service tools can be found in System Service Manual, 2211222-100.

- a) Aperture Test
- b) Collimator Functional Diagnostic
- c) Interconnect Test

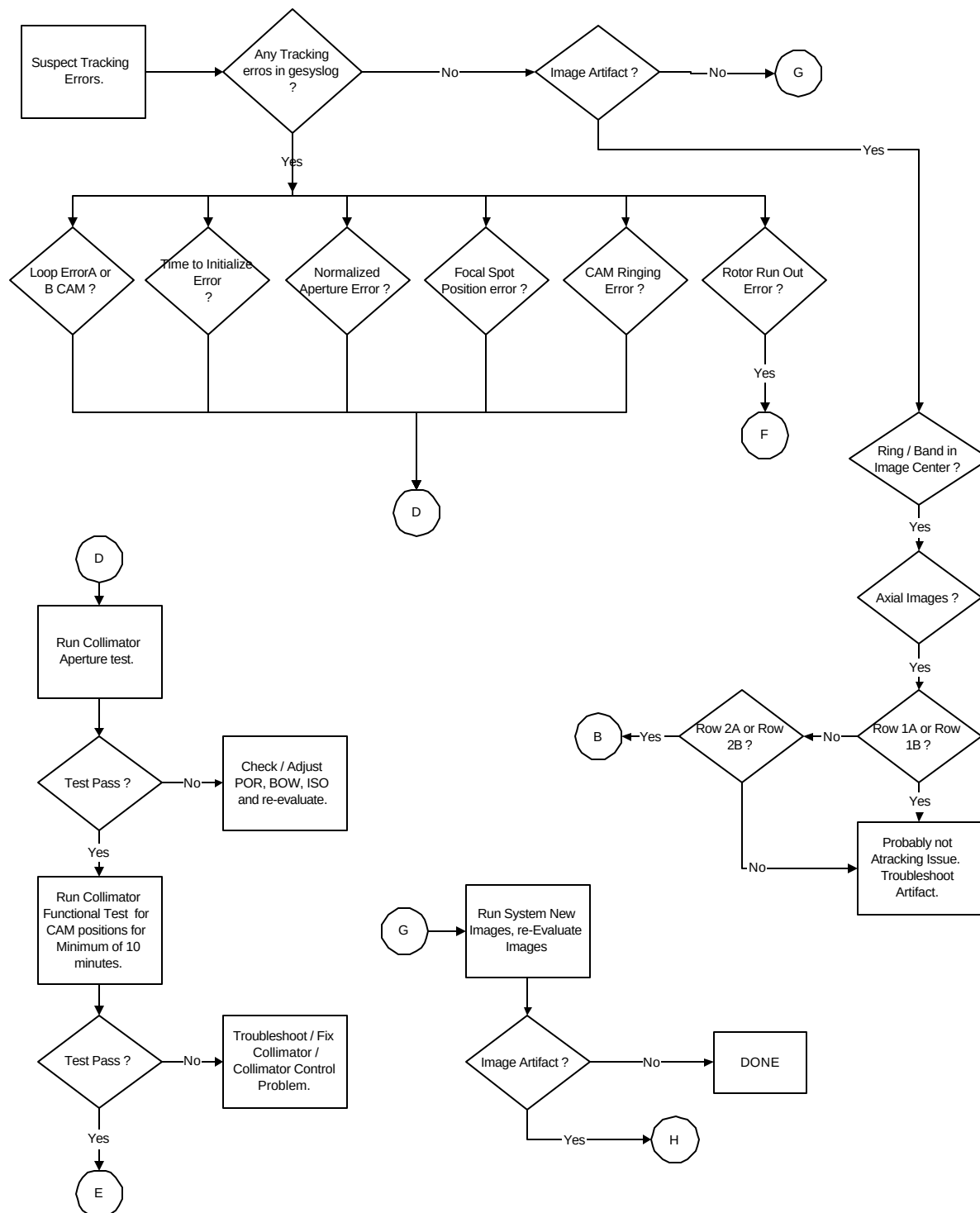
4) Escalate Problem to the On-Line Center and/or CT Engineering.

Byron Matthews (414) 785-4245 Pager: (414) 557-2637

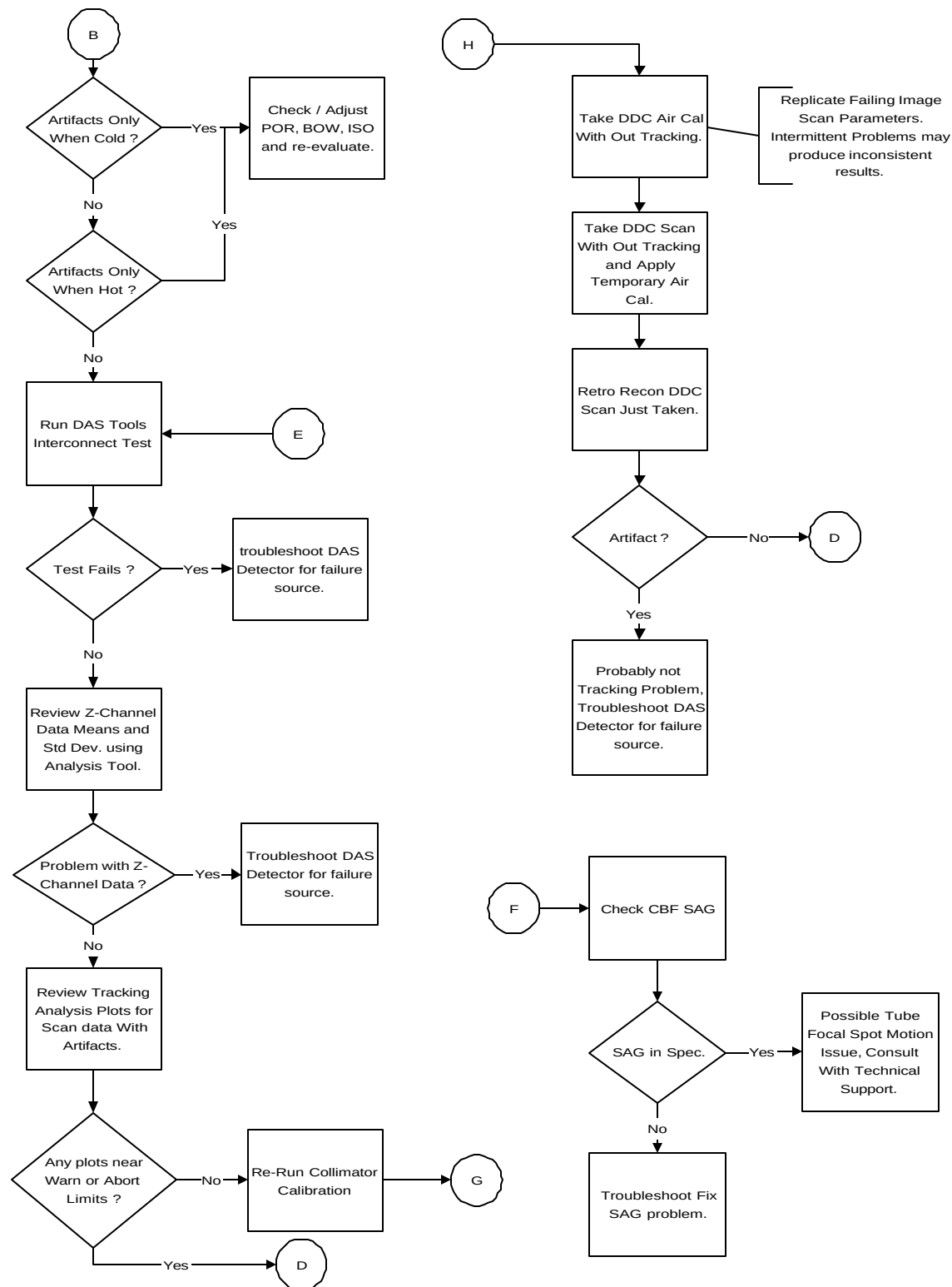
5) Re-load revision 1.0 software. If there are other errors, continue below.

If there are other errors in file or in System Error Log,

1. Stop! Do not continue with Detail Cal. Some Tracking errors can be the result of many factors, such as alignments (POR, BOW), DAS noise issues, Collimator, etc...
1. Follow Tracking troubleshooting flow diagram.
2. Escalate Problem to the On-Line Center and/or CT Engineering.

X-Ray Beam Tracking Troubleshooting Flowchart

X-Ray Beam Tracking Troubleshooting Flowchart continued



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